Day 41 : useState Hook

1. The Fundamental Problem

Question: How do you make a webpage interactive?

Answer: When data changes, the UI must change.

Example:

- User clicks a button → Counter should increase
- User types in input → Text should appear on screen
- User toggles a switch → Theme should change

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2. Why Normal Variables Don't Work

```
function Counter() {
  let count = 0;  // Normal variable

  return (
    <button onClick={() => {
      count = count + 1;
      console.log(count);  // Logs: 1, 2, 3...
    }}>
      Count: {count}  {/* Always shows 0 on screen! */}
    </button>
  );
}
```

What happens:

1. Click button → `count` becomes 1 in memory
2. Console shows `1` ✅

3. But UI still shows `0` ❌

Why?

- React doesn't know the variable changed
- Component function ran once, returned JSX, done
- To update UI, the component must **run again** (re-render)
- Normal variables can't trigger re-renders

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3. What useState Does

`useState` gives you two things:

```
const [count, setCount] = useState(0);
//    ↑      ↑          ↑
//  value  updater    initial value
```

1. `count` - Current value of the state
2. `setCount` - Function to update the value AND trigger re-render
3. `useState(0)` - Sets initial value to 0

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4. How useState Works (Step by Step)

```
function Counter() {
  const [count, setCount] = useState(0);

  return (
    <button onClick={() => setCount(count + 1)}>
      Count: {count}
    </button>
  );
}
```

First Render:

1. `useState(0)` runs → React stores `0` internally
2. Returns `[0, setCount]`
3. `count = 0`
4. JSX: `<button>Count: 0</button>`
5. React renders button showing "Count: 0"

User Clicks Button:

1. `setCount(count + 1)` called → `setCount(1)`
2. React updates stored value from `0` to `1`
3. React schedules a re-render
4. **Component function runs again** (entire function from top)
5. `useState(0)` runs again, but this time returns `[1, setCount]`
6. `count = 1`
7. JSX: `<button>Count: 1</button>`
8. React updates DOM
9. Button now shows "Count: 1"

Key insight: The component function runs again, but `useState` remembers the updated value.

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5. Where Does React Store State?

```
// Simplified: What React does internally

let stateStorage = [];  // React keeps state here (outside your component)
let currentIndex = 0;

function useState(initialValue) {
  const index = currentIndex;
```

```
  // First time: initialize
  if (stateStorage[index] === undefined) {
    stateStorage[index] = initialValue;
  }

  const setState = (newValue) => {
    stateStorage[index] = newValue;  // Update stored value
    reRenderComponent();  // Tell React to re-render
  };

  currentIndex++;
  return [stateStorage[index], setState];
}
```

Key points:

- State lives **outside** your component function
- State **persists** between renders
- Each component instance has **its own state**

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6. Re-render Means Running the Whole Function Again

```
function Counter() {
  console.log("Component function running!");

  const [count, setCount] = useState(0);

  const double = count * 2;  // This is recalculated every render

  return (
    <div>
      <p>Count: {count}</p>
      <p>Double: {double}</p>
      <button onClick={() => setCount(count + 1)}>Increment</button>
    </div>
```

```
  );
}
```

Every click:

1. `setCount` called
2. Entire `Counter()` function runs again
3. Console logs "Component function running!"
4. `double` is recalculated
5. New JSX is created
6. React updates only what changed in the DOM

Important: All variables inside the function are **recreated** every render. Only state values persist (because React stores them externally).

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7. State Updates Are NOT Immediate

```
function Counter() {
  const [count, setCount] = useState(0);

  function handleClick() {
    console.log("Before:", count);  // 0

    setCount(count + 1);

    console.log("After:", count);   // Still 0! (not 1)
  }

  return <button onClick={handleClick}>Count: {count}</button>;
}
```

Why `count` is still 0 after `setCount`?

- `setCount` **schedules** an update, doesn't execute immediately

- Your function must finish first
- Then React processes the update and re-renders
- In the next render, `count` will be 1

State updates are asynchronous!

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8. Multiple setState Calls - Common Mistake

❌ Wrong Way:

```
function handleClick() {
  setCount(count + 1);  // setCount(0 + 1) = setCount(1)
  setCount(count + 1);  // setCount(0 + 1) = setCount(1)
  setCount(count + 1);  // setCount(0 + 1) = setCount(1)
}
// Result: count becomes 1 (not 3!)
```

Why? During the function, `count` is still 0. All three calls are `setCount(1)`.

✅ Correct Way (Updater Function):

```
function handleClick() {
  setCount(prevCount ⇒ prevCount + 1);  // 0 + 1 = 1
  setCount(prevCount ⇒ prevCount + 1);  // 1 + 1 = 2
  setCount(prevCount ⇒ prevCount + 1);  // 2 + 1 = 3
}
// Result: count becomes 3 ✅
```

Rule: When new state depends on old state, use updater function.

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9. Different Types of State

Numbers:

```
const [count, setCount] = useState(0);
setCount(5);
setCount(count + 1);
```

Strings:

```
const [name, setName] = useState('');
setName('John');
```

Booleans:

```
const [isOpen, setIsOpen] = useState(false);
setIsOpen(true);
setIsOpen(!isOpen);  // Toggle
```

Arrays:

```
const [items, setItems] = useState([]);

// Add item
setItems([...items, newItem]);

// Remove item
setItems(items.filter(item => item.id !== idToRemove));

// Update item
setItems(items.map(item =>
  item.id === idToUpdate ? { ...item, name: 'New Name' } : item
));
```

Objects:

```jsx
const [user, setUser] = useState({ name: '', age: 0 });

// Update one property
setUser({ ...user, name: 'John' });

// Update multiple properties
setUser({ ...user, name: 'John', age: 25 });
```

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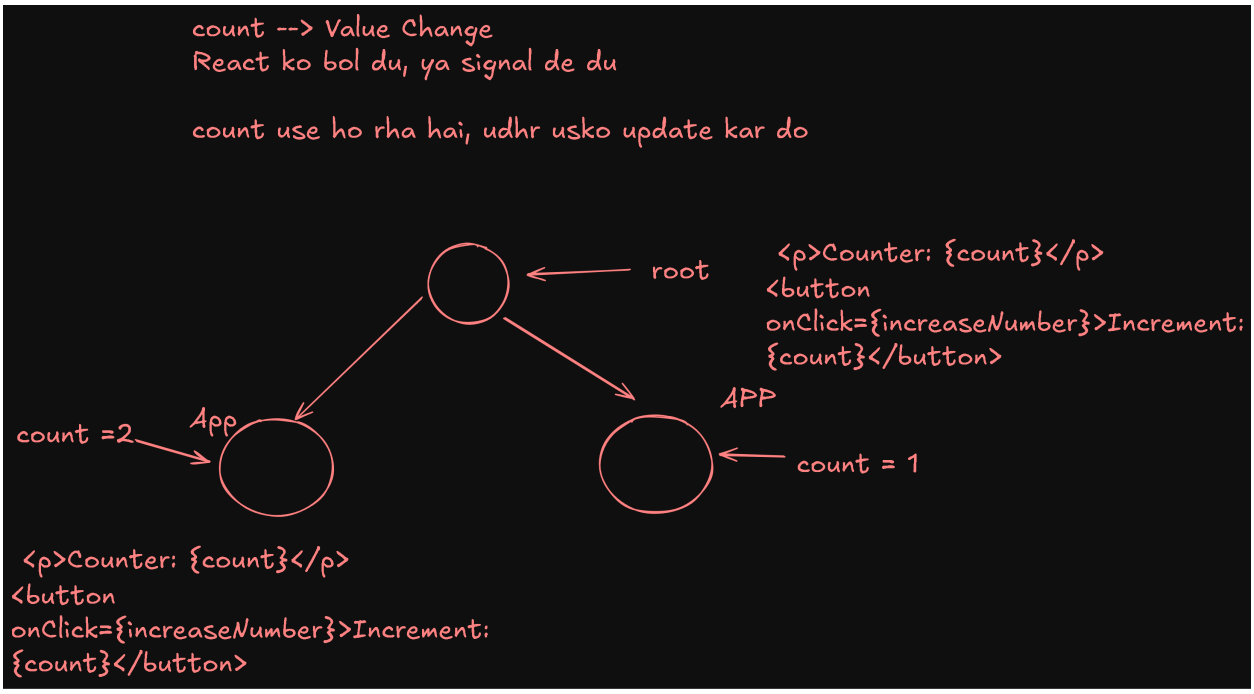
10. Multiple State Variables

```jsx
function Form() {
  const [name, setName] = useState('');
  const [email, setEmail] = useState('');
  const [age, setAge] = useState(0);

  return (
    <div>
      <input value={name} onChange={e => setName(e.target.value)} />
      <input value={email} onChange={e => setEmail(e.target.value)} />
      <input value={age} onChange={e => setAge(e.target.value)} />
    </div>
  );
}
```

You can have as many state variables as you need!


count --> Value Change
React ko bol du, ya signal de du
count use ho rha hai, udhr usko update kar do
root
Counter: {count}
<button
onClick={increaseNumber}>Increment:
{count}</button>
APP
count = 1
App
count =2
Counter: {count}
<button
onClick={increaseNumber}>Increment:
{count}</button>